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Ecological Modelling 237–238 (2012) 120–131

Contents lists available at SciVerse ScienceDirect

Ecological Modelling

journal homepage: www.elsevier.com/locate/ecolmodel

Incorporating canopy gap-induced growth responses into spatially implicit growth model projections

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ARTICLE INFO

Article history:
Received 31 August 2011
Received in revised form 31 March 2012
Accepted 2 April 2012

Keywords:
Forest Vegetation Simulator
Edge effects
Natural disturbance-based silvicultural systems
Intra-stand variability
Canopy gaps
Acadian Forest Ecosystem Research Program

ABSTRACT

Public land management across North America now incorporates multiple ecological and social values and has led to use of increasingly complex silvicultural systems, such as those designed to emulate natural disturbance regimes, in an effort to manage for this wider variety of objectives. In the eastern United States and Canada, canopy gap-based silvicultural systems are often used to promote and sustain intra-stand variability in temporal and spatial patterns. These are difficult to model in many of the region's growth and yield models, with the inherent intra-stand variability tenuously assumed to have negligible effects on stand responses despite increasing evidence to the contrary.

Using the University of Maine's Acadian Forest Ecosystem Research Program (AFERP) as a test case, we investigated this assumption for canopy gap-based systems by first modeling the proportion of area potentially influenced by gaps in these systems with a discrete space model, and then estimating the influence of canopy gap-induced growth increases on growth projections of a calibrated version of the Northeast variant of the Forest Vegetation Simulator (FVS-NE), a spatially implicit model commonly used by managers in this region. Results from the discrete space model showed that, across both AFERP gap-based treatments, up to 52.4% of the stand area can be within one mature tree height of a gap edge, and that the edge area could persist for multiple cutting cycles (>10 years), potentially having a dramatic impact on stand growth and yield. However, results from FVS-NE suggested the contrary, with a 20–50% increase in growth within the edge area only increasing merchantable sawtimber yields by 1–5% over the course of an entire 100-year rotation.

We conclude that edge effects are likely being severely underestimated due to limitations in the FVS model, most notably the relatively inflexible control of stand density index-based growth maximums. Short-term projections with plots not approaching these maximums suggest that a 20–50% edge effect could increase merchantable yields by 15–37% in these gap-based systems, suggesting the need for significant reworking of the underlying FVS model.

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1. Introduction

Sustainable forest management requires information regarding long-term stand dynamics in order to evaluate and compare long-term development under different management alternatives and silvicultural systems. The recent shift from sustained-yield harvesting (Seymour et al., 2006) towards prioritizing forest health and biodiversity (Attiwill, 1994; Franklin and Forman, 1987; Lindenmayer et al., 2006) has stimulated interest in alternative management strategies (Landres et al., 1999; Seymour et al., 2002) and silviculture systems (Keeton, 2006; de Montigny, 2004; Seymour et al., 2006). These strategies are generally coarse-filter approaches to biodiversity management and designed to maintain forest complexity and biodiversity, however defined, within the range of natural variability.

Many current growth and yield models are spatially implicit tree- or stand-level models, a consequence of the historically greater focus on the development of even-aged stand models in comparison to models appropriate for more complex forests (Peng, 2000). Spatially implicit competition indices have been found adequate in predicting growth of stands with uniform regeneration and density (Boivin et al., 2010; Filipescu and Comeau, 2007; Lorimer, 1983); conditions typically created when stands are managed with traditional silviculture systems homogeneously applied over space.

Silvicultural regimes with an emphasis on forest diversity and structural complexity are becoming increasingly complicated as they are implemented heterogeneously over time and space. Several studies have demonstrated that understory light conditions can be manipulated through differences in silvicultural system design such as harvesting intensity, gap size, shape, density,

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http://dx.doi.org/10.1016/j.ecolmodel.2012.04.003

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location and orientation of openings, and spatial pattern and species composition of residual trees, thereby influencing understory survival and growth (Battaglia et al., 2002; Coates et al., 2003; Menard et al., 2002; Sprugel et al., 2009). Furthermore, a growing number of studies suggest changes at the local scale are important in predicting growth at the tree-level (Arseneault et al., 2011; Pedersen and Howard, 2004; Puettmann et al., 2009; York and Battles, 2008). The impacts of dismissing intra-stand variability in stand dynamic responses from canopy gaps during long-term stand dynamics projections are becoming an increasingly relevant concern.

The study of edge dynamics and fragmentation patterns over time in response to forest management has been a focus of landscape ecology, particularly through the use of forest simulation models (Scheller and Mladenoff, 2007). There are many studies regarding effects of management strategy design on spatial configuration, fragmentation, and levels of forest interior over time (Franklin and Forman, 1987; Radeloff et al., 2006; Zheng and Chen, 2000). There is now a need to replicate these types of landscape level investigations on spatial patterns at scales relevant to growth and yield modeling of complex forests to better evaluate alternative silvicultural system designs and their impacts on stand dynamics.

Currently, no single forest model is suitable to meet the practical needs of a forest manager trying to simulate stand dynamics under these complex management regimes. The use of spatially implicit forest growth models to simulate forest dynamics when managed with these regimes is becoming increasingly tenuous given that spatial competition metrics are considered more appropriate and can improve model performance in complex forests (Biging and Dobbertin, 1992; Boivin et al., 2010; Stadt et al., 2007). However, many spatial forest models (e.g., gap models such as JABOWA – Botkin et al., 1972; SILVA – Kercher and Axelrod, 1984; and SORTIE – Pacala et al., 1993) either: (1) have an emphasis on forest succession and lack important practical outputs required for forest management; (2) have limited flexibility in specifying complex harvesting regimes; (3) have data requirements beyond what is traditionally available to forest managers rendering them difficult to calibrate; and/or (4) are intended for use at spatial resolutions too coarse to evaluate alternative silvicultural system designs and their effects on stand level growth and yield (e.g., landscape level models such as LANDIS – He et al., 1999; and LANDIS-II – Scheller et al., 2007). Consequently, spatially implicit models are still used to simulate dynamics in stands managed with complex silvicultural regimes and differences in intra-stand growth responses arising from differing implementations of quantitative prescriptions are assumed to be negligible.

Our goal was to investigate the potential of modifying a spatially implicit growth model to predict stand dynamics under these complex, gap-based silvicultural regimes. We chose to model systems used in the University of Maine's Acadian Forest Ecosystem Research Program (AFERP). The AFERP is testing the use of an irregular group shelterwood and an expanding group selection, where groups expand in alternate cutting cycles (i.e., two gap cohorts; Arseneault et al., 2011). More specifically, our objectives were to: (1) investigate differences in proportions of forest influenced by canopy gaps over time, how long these edge conditions persist before being harvested, and what combinations of harvesting scenarios are likely to occur under the variants of the expanding-gap silvicultural system implemented at the AFERP; (2) integrate evidence of intra-stand variation in spatially implicit stand dynamic growth responses from canopy gaps through data partitioning (i.e., differentially simulating growth under the plausible harvesting scenarios identified during the spatial simulations); (3) conduct a sensitivity analysis by modifying edge forest growth rates between 0% and 50% above the matrix growth rates to evaluate implications of not accounting for intra-stand variation in growth dynamic responses when simulating long-term growth and yield dynamics of these alternative silvicultural systems.

We used Forest Vegetation Simulator (FVS) model for this effort because it is a distance-independent, tree-level growth and yield model that is widely used by government agencies, industry, educational institutions and private landowners in North America (Crookston and Dixon, 2005; Dixon, 2002). This prevalence is in no small part due to the ongoing technical support and user training by the USDA Forest Service. A variety of support tools are available to simplify simulation preparations and output interpretation (e.g., Suppose – Crookston, 1997; SVS – McGaughey, 1997) which make FVS technically accessible to a variety of audiences. Furthermore, its continual development in expanding model capabilities to meet the needs of the forest modeling community (e.g., Fire and Fuels Extension – Rebain et al., 2009; Reinhardt and Crookston, 2003; Climate change extension – Crookston et al., 2010) make this suite of software tools one of the few forest growth models with the flexibility and potential for continued widespread use as demand for modeling these alternative silvicultural systems increases (Monserud and Robinson, 2003).

2. Methods

2.1. Study area and design

The AFERP is located near the towns of Bradley and Eddington, Maine (44°51′N, 68°37′N) and lies within the Penobscot Experimental Forest (PEF), co-managed by the University of Maine and the U.S. Forest Service. The PEF is within the Acadian Forest Region, an ecotone between the northern boreal and southern broadleaf forests with a cool and humid climate. Topographic relief at the PEF is low, with slopes generally less than 20%. Soils are derived from Wisconsian glacial till and range from well drained loams and sandy loams on glacial till ridges to poorly and very poorly drained loams and silt loams in flat areas between the ridges (Saunders and Wagner, 2008; Sendak et al., 2003). Forest cover types in the PEF are diverse, ranging from nearly pure coniferous overstories dominated by red spruce (*Picea rubens* Sarg), balsam fir (*Abies balsamea* (L.) Mill.), eastern white pine (*Pinus strobus* L.), and eastern hemlock (*Tsuga canadensis* (L.) Carrière); to mixedwood overstories consisting of those same conifers and hardwoods such as sugar and red maple (*Acer saccharum* Marsh. and *Acer rubrum* L., respectively), paper and gray birch (*Betula papyrifera* Marsh. and *B. populifolia* Marsh., respectively), and trembling and largetooth aspen (*Populus tremuloides* Michx. and *Populus grandidentata* Michx., respectively) (Saunders and Wagner, 2008). A complete list of trees species found on the AFERP sites along with their respective shade tolerance rankings is provided in Table A.1.

The AFERP study includes three treatments replicated three times across nine, 8.9–11.3 ha research areas (RA) within irregularly-aged mixedwood sites. Treatments consist of an experimental control and two expanding-gap systems based on the German "Femelshlag" system: (1) an extended group shelterwood with reserves (hereafter referred to as "large-gap"), with 20% of the area removed on a 10-year cutting cycle and all removals occurring during the first half of the 100-year rotation and (2) an "expanding" group selection (hereafter referred to as "small-gap") with 10% area removal on a 10-year cutting cycle. In the latter treatment, gaps are divided into two cohorts that are harvested during alternate expansion/cutting cycles throughout the 100-year rotation. More detailed descriptions of the AFERP's design and inventory systems can be found in Saunders and Wagner (2005).

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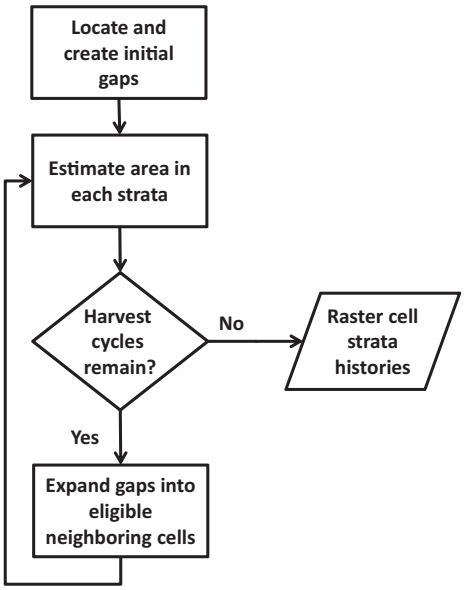

Fig. 1. A simplified flowchart of the discrete space model and output to inform FVS-NE simulations. A more detailed flowchart can be found in the appendix.

2.2. Data

Pre-treatment measurements were conducted in 1995–1997, with post-treatment inventories conducted every five years starting in 2000–2002. A nested sampling design was used, with all overstory trees ≥9.5 cm diameter at breast height (DBH) measured within 0.05 ha circular sample plots and saplings between 1.5 cm and 9.5 cm DBH measured within 0.01 ha circular subplots centered within the overstory plots. Prior to both initial harvest and gap delineation, 20 overstory plots were initially installed per replicate by randomly selecting locations from intersections of a 50 m × 50 m grid overlain on the replicate.

2.3. Discrete space model and analysis

Changes in forest strata over time were studied under various expanding-gap harvest systems through the use of a raster-based simulation model (Fig. 1) created in R version 2.11.1 (R Development Core Team, 2011) with the use of the raster (version 1.7-29 – Hijmans and van Etten, 2011) and sp (version 0.9-69 – Bivand et al., 2008) packages. During each simulation, a 9 ha stand (300 m × 300 m) was harvested based on rules designed to mimic the expanding-gap silviculture systems within AFERP. Ten initial harvest gap centroids were generated using a random sample of evenly distributed grid points to assure gaps were not overlapping. Centroid locations were deviated from the grid points with random noise to better resemble actual harvest patterns. For both treatments, initial gap cohorts were harvested as square openings of 900 m^2 (30 m × 30 m) and 1600 m^2 (40 m × 40 m with feathered edges to achieve 1800 m^2) to represent the 1% and 2% gap area targets of the small-gap and large-gap treatments, respectively. On-the ground application of the AFERP treatments has ranged from concentric expansion of gaps (i.e., 4-side expansion) to asymmetrical expansion on only two sides. Therefore, both gap-expansion extremes were modeled in this study, and the mean of those extremes was used to better approximate gap edge dynamics. Gap expanding harvests were iteratively implemented. At the beginning of each harvest cycle, the total area to be harvested was equally divided amongst all gaps eligible for expansion. A simple search algorithm was then used to identify all cells adjacent to existing gap cells eligible for harvest. During asymmetrical expansions, cells in the specified direction were given priority for harvest. If a gap's harvest allocation could not be met through harvest of the prioritized cells alone, it was allowed to randomly harvest any eligible cells until its harvest allocation was met. If no adjacent cells were eligible for expansion, gap expansion stopped and any remaining harvest quota was equally redistributed to the remaining gaps capable of expanding.

After each simulation cycle, an output raster was created where each cell represents 100 m^2 (or approx. 0.1% of the total stand surface area) and contains its state during that cycle (i.e., matrix, edge, or gap). Edge was differentiated from matrix if cells fell within 20 m of a gap cell, following the work of Arseneault et al. (2011) who found evidence of increased tree growth within these systems if plots fell within one tree height of a gap edge. A total of five or ten simulation cycles were implemented for the large gap and small gap treatments, respectively.

Output from the discrete space model was a listing of the proportion of cells within the raster that had the same strata history at each simulation cycle. The model was run 1000 simulations per treatment and gap expansion combination, i.e., 4000 total, to better approximate mean raster cell histories. Cell histories that were deemed unlikely to occur during implementation (<1% and <0.1% of the area in the large-gap and small-gap treatments, respectively) were eliminated. Proportions of these flagged histories were reallocated to the remaining histories combinations that occurred during the same period proportional to their occurrence in order to ensure the harvest quota was met during each harvest cycle (see Arseneault, 2011; Table A.2). Lastly, the proportion of raster area in each stratum, summed across all cell histories, was calculated for each model run. Confidence bounds were estimated using simulation sample quantiles generated using the stats (version 2.12.1 – R Development Core Team, 2011) package in R.

2.4. FVS growth model and analysis

All growth simulations were done using the test version of the northeast variant of FVS (FVS-NE [release version: 2/16/2011]) and the database extension (Crookston et al., 2005) to manage data input and output using the Microsoft Access® 2010 relational database management system (Microsoft Corporation, Redmond, USA). FVS is a collection of growth and yield models specifically calibrated to forest types within a given region (i.e., variants) with numerous user customizations to the model (i.e., keywords) available to adjust model predictions, simulate management activities, etc. This modeling framework has been described extensively elsewhere in the literature (e.g., Crookston and Dixon, 2005; Dixon, 2002) and online (http://www.fs.fed.us/fmsc/fvs/). Although variants are in various stages of development and differ slightly, all require the bare minimum measurements of tree diameter and tree species, along with information on inventory design and site characteristics. From the initial data, stand level maximums are defined for either stand density index (SDI; Reineke, 1933) or basal area, usually calculated as a weighted average of the variant-specific, pre-defined maximums of the initial species composition within the data. Simulations then proceed with thinnings, harvests, and other management activities by adjusting the tree counts that each record in the data. The remaining trees are then grown with either large-tree diameter or small-tree height growth models, with the converse variable estimated from species-specific allometric relationships. Density-independent mortality is then estimated to adjust tree counts. At this point, stand-level parameters are estimated; if stand exceeds the predefined basal area or SDI maximum, density-dependent mortality is used to adjust tree counts again, usually with a function that preferentially removes the smallest and/or most shade intolerant trees. Regeneration may be added to the tree records at this point either through user input

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or, in a few variants, a full-establishment regeneration sub-model. Summary statistics for that cycle are then calculated and output before the next cycle begins (Dixon, 2002). For specific details on the FVS-NE program (e.g., diameter and height growth functions, mortality estimators, etc.), refer to Dixon and Keyser (2008).

Using a 10-year cutting cycle, stand development over a 100-year rotation was projected for each harvest scenarios, i.e., raster cell histories, output by the discrete space model. FVS-NE simulations were run at the plot level (i.e., plots were treated as individual stands) to allow for the incorporation of variable stand characteristics and enable the integration of intra-stand variety in growth response. Variation in site characteristics were included by estimating the balsam fir site index$_{50}$ for every plot using local estimates of depth to water table and the site-classification system from Briggs (1994), which were then scaled to represent a moderately productive site with a mean of 19.8 m. FVS-NE currently does not have a full-establishment model for regeneration. Therefore, initial and post-harvest regeneration composition and densities were estimated based on empirical data. Because AFERP is still in its early phases, the long-term effects of these expanding-gap harvests on regeneration establishment, survival, and recruitment are unknown. Regeneration inputs were, therefore, generated in proportion to initial seedling and sapling importance values in order to be representative of pre-treatment composition. Sapling importance values (IV) were sums of relative dominance (%), relative frequency (%), and relative density (%), while seedling IVs were sums of relative frequency and relative density.

Initial regeneration records were added to the input tree list with an approximately 1.0 cm (0.4 in.) diameter and limited to a maximum of 4448 trees per ha (1800 trees per acre) to minimize bias in projected basal area growth and mortality based on recommendations in Ray et al. (2009). Based on post-harvest observations, regeneration survival (i.e., all stems <10.2 cm DBH) during harvesting was assumed to be 80%, which is within the ranges reported in Stokes et al. (2009).

Given that the regeneration predominantly consisted of shade tolerant conifers and hardwoods, the basal area maximum was set at 46.75 m^2 ha^{-1} (203.7 ft^2 ac^{-1}), which corresponded to the 85th percentile of the pre-treatment basal area distribution across all plots. This level is representative of mature stand conditions in these stand types (Saunders and Wagner, 2008). The large tree diameter growth model was calibrated with the empirical AFERP data using a truncated diameter growth distribution to protect against bias due to measurement errors. Diameter growth increments were estimated as the difference between the first two subsequent diameter at breast height measurements for all live trees in the matrix plots (i.e., 10 years of growth). Negative increments were entered as 0, and increments greater than the 95th percentile by species were capped at the 95th percentile. The small tree height model was not calibrated due to lack of data.

The increased growth effect in edge conditions for large trees and small trees were adjusted using the BAIMULT and REGHMULT keywords. These multipliers correspond to the primary growth submodels for large and small trees, respectively, and increase growth rates without modifying underlying height/diameter ratios. Multipliers of 0%, 10%, 20%, 25%, and 50% were used to evaluate the potential impact of growth increases in edge conditions. These levels were representative of the increased growth rates in canopy gap-created edge observed in Arseneault et al. (2011; 25–60%) and Banal et al. (2007; 30–66%). Since edge was defined as a function of distance to harvest boundaries, the edge designation was purely a measure of gap adjacency and does not account for the effects of canopy gap areas transitioning back into a forested condition over time. The duration of edge growth rate increases was based on studies of growth following tree release during precommercial and commercial thinnings (Leak, 2007; MacLean and Morgan, 1983; Pothier, 2002; Weiskittel et al., 2009). Duration of increased growth rates vary but have been observed to last between 10 and 20 years after release. Consequently, the full value of these multipliers were added during the first cycle a plot transitions into the edge condition, and were halved if the condition persisted through a second 10-year cycle, after which the multiplier was removed (Fig. A.2). Harvesting rules were specified to emulate the guidelines used during the AFERP harvests. In both systems, reserve trees of long-lived (e.g., white pine) or uncommon species (e.g., red oak, large sugar maple) were retained in gaps at a basal area density of 3.7 m^2 ha^{-1} (16.1 ft^2 ac^{-1}), approximately 10% of the pre-harvest stocking. Low-quality, short-lived species were targeted for removals; these included aspen, birch, fir and red maple. FVS outputs included plot density and basal area over time, as well as final diameter distribution in 5 cm (2 in.) diameter classes for shade tolerant and intolerant conifers and hardwoods based on shade tolerance reported in (USDA, 1990; Table A.1).

To summarize stand-level responses by treatment, a Monte Carlo approach was used to combine the FVS-NE plot-level output with the raster cell histories, i.e., harvesting scenarios, obtained from the discrete space model (Table A.2). For each treatment, observed proportions of likely harvesting scenarios were transformed to plot counts (1 plot/175 total plots ≈0.57%) and 100 replicates were created by sampling, with replacement, the appropriate number of plots from each scenario. Raw output data was processed and summarized at the stand level in the Microsoft Access® relational database management system and using the plyr package in R (version 1.4 – Wickham, 2011). Confidence bounds were estimated with the Monte Carlo replicate quantiles generated using the stats package (version 2.12.1 – R Development Core Team, 2011) in R and figures were produced using the ggplot2 package (version 0.8.9 – Wickham, 2009).

Lastly, to investigate differences in the FVS-NE model behavior and better elucidate effect of increased growth rates on stand dynamics, FVS-NE output from two subsets of plots were chosen, one set near the maximum sustainable basal area when growth multipliers were applied (hereafter referred to as "40 BA plots", BA = 42.2 ± 0.7 m^2 ha^{-1}, n = 9), and a second subset with a lower basal area (hereafter referred to as "30 BA plots", BA = 28.5 ± 0.7 m^2 ha^{-1}, n = 10). Both subsets of plots transitioned into edge for the first two harvest cycles, and were harvested during the third cycle. These data were summarized using the same Monte Carlo approach as outlined above.

3. Results

3.1. Edge simulation model

The large-gap design scheduled all harvesting during the first half of the rotation which caused a rapid reduction in matrix and edge strata in comparison to the small-gap treatment, in which these strata declined gradually over the rotation (Fig. 2). When examined over an entire rotation, the small-gap treatment created more area in an edge condition than did the large-gap treatment (95% confidence interval [CI$_{0.95}$]: 73.5–79.0% and 60.0–73.5%, respectively). Proportions of forest in an edge condition were greatest during the initial harvest cycles, and were greater in the small-gap treatment (CI$_{0.95}$: 42.7–52.4%) than the large-gap treatment (CI$_{0.95}$: 27.3–38.8%).

Simulations of the small-gap treatment often did not harvest the entire stand by the end of the harvest rotation (CI$_{0.95}$: 96.7–100% of the total area harvested; Fig. 2), an artifact caused by the algorithm used to simulate alternating harvesting of gap cohorts with random gap placement. Relaxing model rules concerning the rigid implementation of harvest quotas and creation of new gaps over

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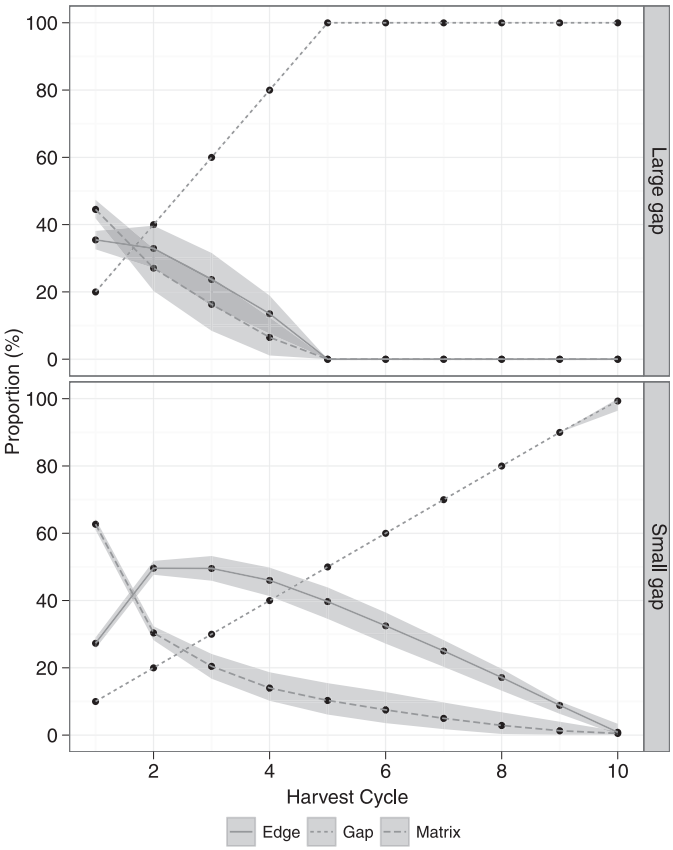

Fig. 2. Changes in proportions of matrix, edge, and gap strata by 10-year harvest cycles for the large- and small-gap treatment. Solid lines indicate mean proportions, while shading indicates the simulated 2.5 and 97.5 percentiles. Darker shading represents overlap between confidence bounds.

time could have eliminated this artifact, but the artifact's effects were deemed minor enough to ignore for FVS simulations.

New edge creation under the large-gap system was limited to the first four harvest cycles, while the alternating small-gap system created new edge throughout the rotation (Fig. 3). The large-gap system created significantly more edge during the first harvest cycle in comparison to the alternating small-gap system ($CI_{0.95}$: 32.3–38.3% and 25.4–29.0%), respectively), while the inverse occurred during the second cycle ($CI_{0.95}$: 10.9–20.2% and 20.2–24.5%, respectively). This difference is a result of the alternate harvesting design, which established a new cohort of gaps within the matrix during the second cycle. There was no significant difference in new edge creation during the third and fourth harvest cycles when both treatments were expanding existing gaps.

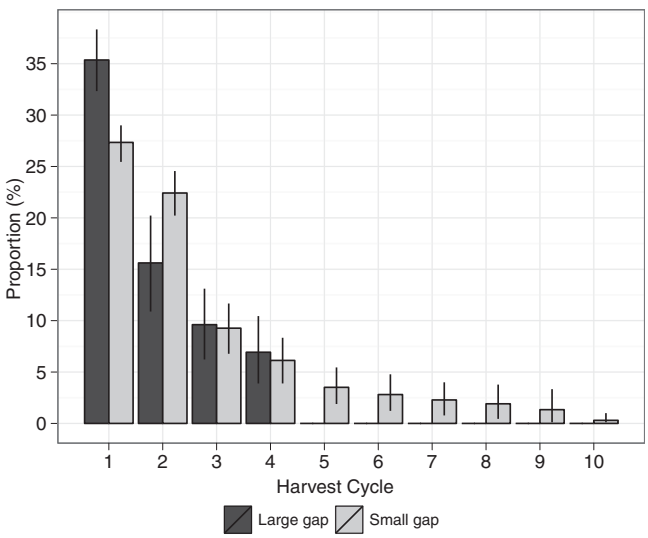

Fig. 3. New edge creation by 10-year harvest cycle by treatment as a proportion of total area. Bars indicate simulated 2.5 and 97.5 percentiles.

Edge longevity generally decreased over time for both treatments as gap expansions began to fully occupy the forested area (Table 1). Overall, the edge stratum was more likely to be harvested quickly in the large-gap treatment in comparison to the small-gap treatment due to alternating harvests between gap cohorts. For example, 51.3% of the edge created during the first harvest cycle of the large-gap treatment was harvested the following cycle in comparison to 1.2% for the small-gap treatment. Consequently, the probability of edge surviving for two or more harvest cycles was greater in the small-gap treatment when it occurred during the first two cycles (98.8% and 98.9%), largely because the establishment of the second gap cohort predominantly consisted of harvesting from the matrix strata.

The peak in short-lived edge conditions occurred early in the rotation for the large-gap treatment and much later in the rotation for the small-gap, when canopy expansions from both cohorts began to significantly harvest portions of their respective edge conditions. For example, 48.2% of the edge stratum that only survived one cycle occurred during the first cycle of the large-gap treatment, compared to the small-gap treatment where 51.5% occurred during the last three harvest cycles.

3.2. FVS-NE simulations

3.2.1. Forest structure

At the stand level, there were differences in projected patterns of forest density (Fig. 4a) and basal area (Fig. 4b) over time across treatments. In the small-gap treatment, the density, basal area and quadratic mean diameter were relatively stable once both gap cohorts had been established. For example, in the 0% edge rate increase, these parameters varied between 1400 and 1880 trees ha^{-1}, 36.2 and 41.0 m^2 ha^{-1}, and 15.8 and 19.0 cm, respectively, for the remainder of the rotation. In the large-gap treatment, structure was relatively stable during the first five harvest cycles, with density, basal area, and quadratic mean diameter varying between 2460 and 3500 trees ha^{-1}, 31.3 and 36.8 m^2 ha^{-1}, and 11.4 cm and 13.5 cm, during this period. During the second half of the rotation, residual forest structure was driven by density dependent mortality resulting in fewer, but larger trees at the end of the harvest rotation. Increasing edge growth rates resulted in lower tree densities and greater basal area regardless of treatment in every harvest cycle. For example, for the large-gap treatment, 20% and 50% edge growth rate increases resulted in mean density reductions up to 7.0 and 14.1% in stem density, respectively, and mean increases in basal area up to 2.3% and 5.6%, respectively. In the small-gap treatment, the 20% and 50% growth rates resulted in relative density reductions up to 5.1% and 10.8%, respectively, and basal area increases up to 1.4% and 3.3%, respectively.

These effects were more pronounced at the plot level (Fig. 5) and decreased as plot basal area approached the maximum sustainable basal area. For example, the second harvest cycle of the 30 BA plots had 20% and 50% edge growth rate increases resulting in mean density reductions of 16.6% and 37.1% and basal area increases of 4.3% and 7.6%, respectively, relative to the 0% increase scenario. In the 40 BA plots, these growth rates resulted in relative density reductions of 14.4% and 35.3% and basal area increases of 3.1% and 4.3%, respectively.

There were differences in species composition by diameter class at the end of the harvest rotation across treatments and edge

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Table 1
Simulated mean, 2.5, and 97.5 percentiles for the proportion of edge longevity by treatment, the harvest cycle during which the edge condition was created (rows) and its subsequent longevity in harvest cycle (columns). Conditional probabilities can be estimated from this table by dividing relevant mean proportions by the appropriate sum (e.g., in the large-gap treatment, to determine the mean probability of edge created during the first harvest cycle being harvested during the second cycle = 18.21/(18.21 + 12.17 + 3.59 + 1.52) = 51.3%).

Treatment	Harvest cycle	Longevity (in 10-year harvest cycles)									
		1	2	3	4	5	6	7	8	9
Large-gap	1	18.21 (17.11, 19.22)	12.17 (8.44, 15.89)	3.59 (1.11, 6.56)	1.52 (0.11, 3.89)					
	2	6.51 (3.56, 9.44)	7.37 (3.67, 11.67)	1.80 (0.11, 4.33)						
	3	6.17 (3.89, 8.56)	3.44 (0.89, 6.11)							
	4	6.93 (3.89, 10.44)								
Small-gap	1	0.34 (0.11, 0.70)	9.24 (8.56, 10)	1.19 (0.33, 2.33)	7.42 (5.33, 9.22)	1.11 (0.22, 2.22)	4.04 (3, 5.11)	1.18 (0.33, 2.11)	2.51 (1.00, 4.00)	0.58 (0.11, 1.67)
	2	0.26 (0.11, 0.67)	8.21 (7.00, 9.22)	0.58 (0.11, 1.44)	6.66 (4.44, 8.78)	0.98 (0.33, 1.89)	3.43 (2.22, 4.78)	0.78 (0.11, 1.56)	1.65 (0.33, 3.33)	0.11 (0.11, 0.11)
	3	0.32 (0.11, 0.78)	1.87 (0.56, 3.44)	0.76 (0.11, 1.62)	3.30 (1.78, 4.89)	0.90 (0.22, 1.67)	1.74 (0.78, 2.78)	0.44 (0.11, 1.11)	0.20 (0.11, 0.56)	
	4	0.19 (0.11, 0.44)	1.29 (0.33, 2.78)	0.49 (0.11, 1.11)	2.65 (1.11, 4.44)	0.60 (0.11, 1.33)	1.07 (0.22, 2.11)	0.12 (0.11, 0.22)		
	5	0.22 (0.11, 0.56)	0.91 (0.11, 2.11)	0.48 (0.11, 1.11)	1.60 (0.56, 2.78)	0.48 (0.11, 1.22)	0.21 (0.11, 0.67)			
	6	0.27 (0.11, 0.78)	0.95 (0.11, 2.00)	0.60 (0.11, 1.44)	1.14 (0.22, 2.33)	0.19 (0.11, 0.56)				
	7	0.33 (0.11, 1.00)	1.26 (0.33, 2.44)	0.74 (0.11, 1.89)	0.29 (0.11, 1.00)					
	8	0.55 (0.11, 1.44)	1.42 (0.22, 2.78)	0.15 (0.11, 0.33)						
	9	1.20 (0.11, 3.04)	0.54 (0.11, 2.22)							

growth rates. The negative exponential distribution in the small-gap treatment was produced over time as regular regeneration influx from harvesting combine with residual trees left during harvests which occurred in previous cycles to grow into the larger diameter classes (Fig. 6a). The tail of the large-gap distribution (45+ cm diameter classes) was largely composed of trees retained during harvests and surviving advance and harvest regeneration from plots harvested early during the rotation (Fig. 6b). Once the last harvest occurred in the fifth harvest cycle, there were no further regeneration events and increasing levels of density dependent mortality targeted shade intolerant trees with low social rank as the stand approached the basal area maximum. Trees in smaller diameter classes then began to die out during the second half of the harvest rotation leaving a peak in density in the 15–30 cm diameter classes composed of surviving regeneration from most recent harvest entries.

Overall, the diameter distribution transitioned to favor shade-tolerant conifers and hardwoods, although intolerant hardwoods are still present in the smaller diameter classes (15–30 cm) due to prolific regeneration. Intolerant species were largely absent from the larger diameter classes, except when retained during harvests (e.g., white pine).

The effect of growth rate multipliers was less pronounced at the stand level (Fig. 6) than the plot level (Fig. 7), but the same trends were observed at both scales. Increased growth rates in edge forests shift projected diameter distributions through two mechanisms: (1) by increasing recruitment into larger diameter classes through increased growth rates of advance regeneration and overstory trees and; (2) hastening onset of density-dependent mortality, which favors mortality of shade intolerant trees with low rank in the height distribution, further contributing to the distribution shift (Fig. 8).

3.2.2. Standing merchantable volume

At the plot level, growth multipliers affected both short-term and long-term patterns of merchantable timber volume (Fig. 7). For example, at the end of the second harvest cycle, edge growth rate multipliers of 20% and 50% increased mean projected volume in standing merchantable pulpwood by 7.8% and 17.5% in the 30 BA plots and 4.3% and 9.2% in the 40 BA plots, respectively. Merchantable sawlog volumes also increased by 14.6% and 35.9% in the 30 BA plots and 7.3% and 16.1% in the 40 BA plots, respectively. Projected standing merchantable volume at the end of the rotation similarly increased with increasing growth rates, and product ratios shifted such that greater proportions of total merchantable volume met the criteria for sawtimber. For example, when edge growth rates and forest matrix rates are equal, approximately 75% and 81% of the total merchantable volume in the last harvest cycle meets the standards for sawtimber in 30 BA and 40 BA plots respectively. When edge growth rates increase by 20% and 50% relative to matrix rates, this proportion increases by 4% (78% of total) and 9% (82% of total), respectively, in 30 BA plots and 1% (82% of total) and 3% (84% of total), respectively, in 40 BA plots.

At the stand level, observed trends were similar to plot level trends, but differences between edge growth rates were less pronounced (Fig. 9). For example, growth multipliers of 20% and 50% increased end of rotation sawtimber yields by 3.5% and 7.5% respectively for the large-gap treatment, and 4.0% and 9.0% respectively for the small-gap treatment relative to projected yields using matrix growth rates.

3.2.3. Merchantable volume harvested

At the plot level, growth multipliers increased both pulpwood and sawtimber merchantable volumes harvested (Fig. 10a). For example, relative to the no edge effect scenario, edge effects of 20% and 50% increased projected mean pulpwood volume

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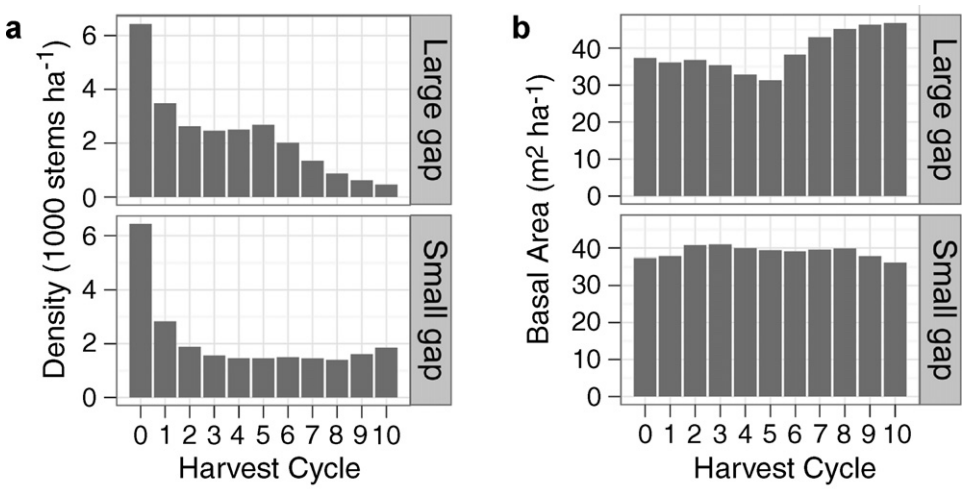

Fig. 4. Mean (a) density (trees ha^{-1}) and (b) basal area (m^2 ha^{-1}) by 10 year harvest cycle for the large- and small-gap treatments assuming no edge growth increases. Error bars for these relationships are not shown; 95% confidence envelopes are consistently less than 1% of the mean values.

harvested 8.4% and 18.8% in the 30 BA plots and 4.2% and 8.8% in the 40 BA plots, respectively. Merchantable sawtimber volumes also increased by 15.2% and 37.9% in the 30 BA plots and 6.9% and 15.4% in the 40 BA plots, respectively.

Stand level trends were similar to those observed at the plot level except differences between edge growth rates were less pronounced (Fig. 10b). For example, growth multipliers of 20% and 50% increased end of rotation sawtimber yields by 2.4% and 5.7% respectively for the large-gap treatment, and 1.8% and 4.0% respectively for the small-gap treatment relative to projected yields under the no edge effect scenario.

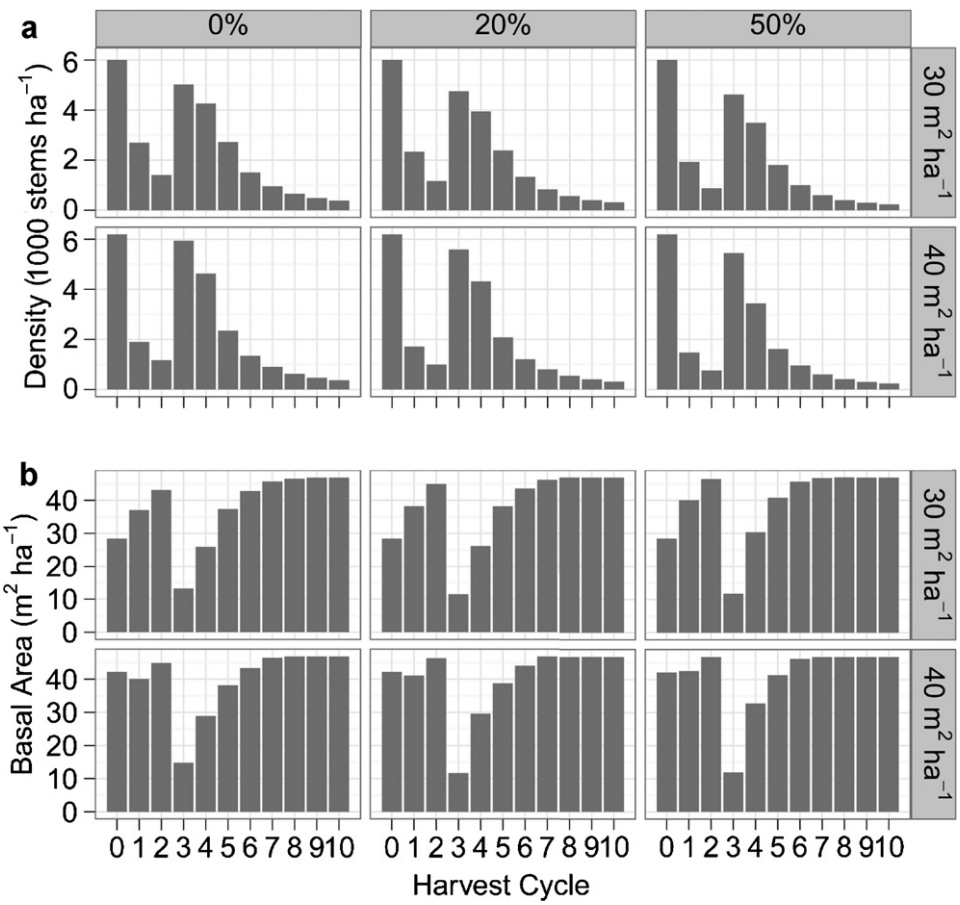

Fig. 5. Mean (a) density (trees ha^{-1}) and (b) basal area (m^2 ha^{-1}) by 10 year harvest cycle for the 30 basal area plots (mean = 28.5 ± 0.7 m^2 ha^{-1}) and 40 basal area plots (42.2 ± 0.7 m^2 ha^{-1}) with increased edge growth rates of 0%, 20%, and 50%. Error bars for these relationships are not shown; 95% confidence envelopes are consistently less than 1% of the mean values.

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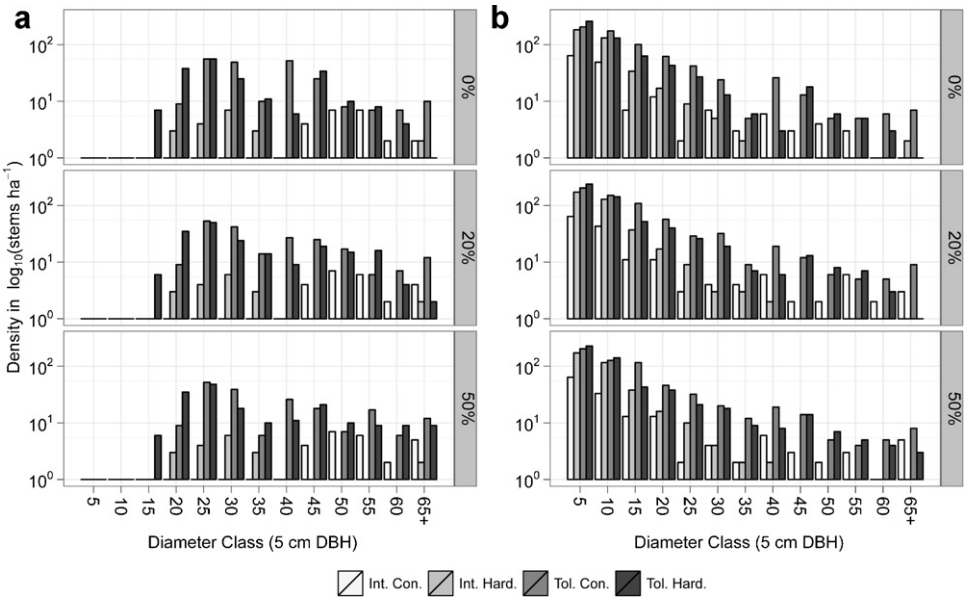

Fig. 6. Diameter distribution by 5 cm diameter classes at the end of the harvest rotation for the (a) large-gap and (b) small gap treatments with increased edge growth rates of 0%, 20%, and 50%.

4. Discussion

4.1. Canopy gap effects on edge creation

The edge simulation model used in this analysis is a simplified representation of the underlying behavior behind the expanding-gap harvesting systems implemented at AFERP. In reality, gap placement and expansions during harvest operations are not randomly allocated nor exactly equal every harvest cycle. Factors such as topographical constraints, spatial distribution and density of advance regeneration, presence of existing harvesting road networks and skid trails, and other silvicultural and practical considerations will influence the design of the harvest system. Additional considerations such as machinery and harvesting crew quality will affect the implementation. Together these factors will affect gap-created edge dynamics.

Nevertheless, the theoretical simulations in this analysis serve as a proof of concept and clearly demonstrated that patterns of edge creation and longevity over time can differ markedly as a result of differences in implementations of complex silvicultural regimes over time and space. In the two expanding-gap system designs implemented at AFERP, the quantities of edge created and their subsequent longevity over time differed significantly. A particularly interesting observation was the effect of alternating harvests between gap cohorts, as implemented in the small-gap treatment, which resulted in increases in early edge longevity and delayed short-lived edge conditions until gap expansions began to fully occupy the stand during the harvest cycles at the end of the rotation. Given the increasing evidence of canopy gap effects on growth

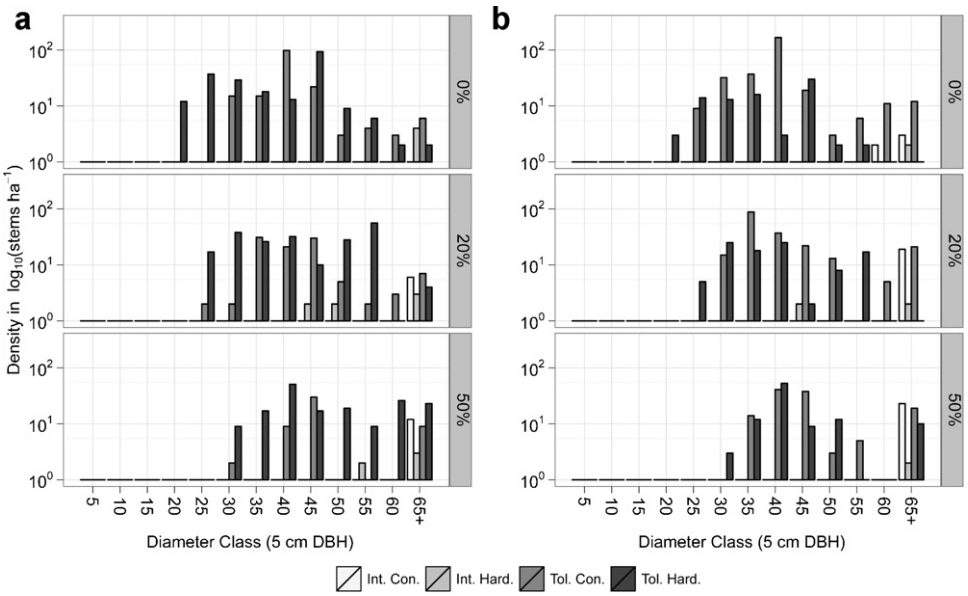

Fig. 7. Diameter distribution by 5 cm diameter classes at the end of a 100-year harvest rotation for the (a) 30 basal area (mean = 28.5 ± 0.7 m^2 ha^{-1}) and (b) 40 basal area (42.2 ± 0.7 m^2 ha^{-1}) plots with increased edge growth rates of 0%, 20%, and 50%. Harvest cycles are each 10 years in length.

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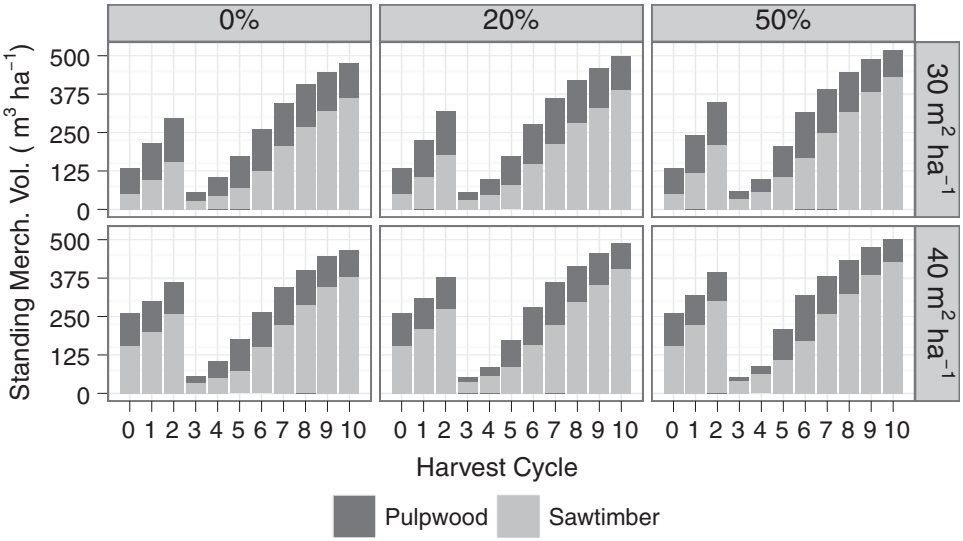

Fig. 8. Mean standing merchantable volume ($m^3\ ha^{-1}$) by 10 year harvest cycle for the 30 basal area plots (mean = 28.5 ± 0.7 $m^2\ ha^{-1}$) and 40 basal area plots (42.2 ± 0.7 $m^2\ ha^{-1}$) with increased edge growth rates of 0%, 20%, and 50%.

dynamics in adjacent edge (Arseneault et al., 2011; Banal et al., 2007; Puettmann et al., 2009), our analysis further demonstrated that failing to incorporate differences in edge creation and longevity patterns when projecting both short- and long-term stand growth and development under these complex silvicultural regimes, could potentially result in severe underpredictions of forest growth and yield. For example, patterns of forest structure and composition over time increasingly diverged as edge growth rates increased relative to matrix growth rates and were greatest in plots that were not near maximum site capacity, such as the subset of 30 BA plots used in our plot-level analysis.

4.2. FVS-NE model limitations

Overall, given the high proportions of edge creation and longevity estimated by the edge simulation model, incorporating edge effects into FVS-NE growth predictions resulted in small differences in stand level outputs over an entire 100-year rotation. However, we believe current limitations of the FVS growth model, which stem from the underlying design philosophy and current model structure, act to dampen the impacts of increasing edge effects at the stand level. Consequently, the importance of including edge effects on long-term stand dynamics projections is being underestimated for several reasons.

First, if sufficient increment data is provided, FVS calibrates the growth models by estimating scale factors for the large-tree diameter growth and small-tree height growth models. The scale factors proportionally adjust the predicted growth increments to better fit observed growth rates and are attenuated over time during long-term projections to approach the base model, which is assumed to be a more stable predictor of growth (Dixon, 2002). Consequently, the effect of growth multipliers on changes in growth increment predictions of individual trees diminishes over time, even if the multiplier itself remains constant. The effect of growth multipliers attenuate because multipliers are applied to proportionally modify increment growth estimates which are predicted as a function of the calibrated scale factors, which also attenuate over time. The results of this behavior is that constant growth multipliers do not

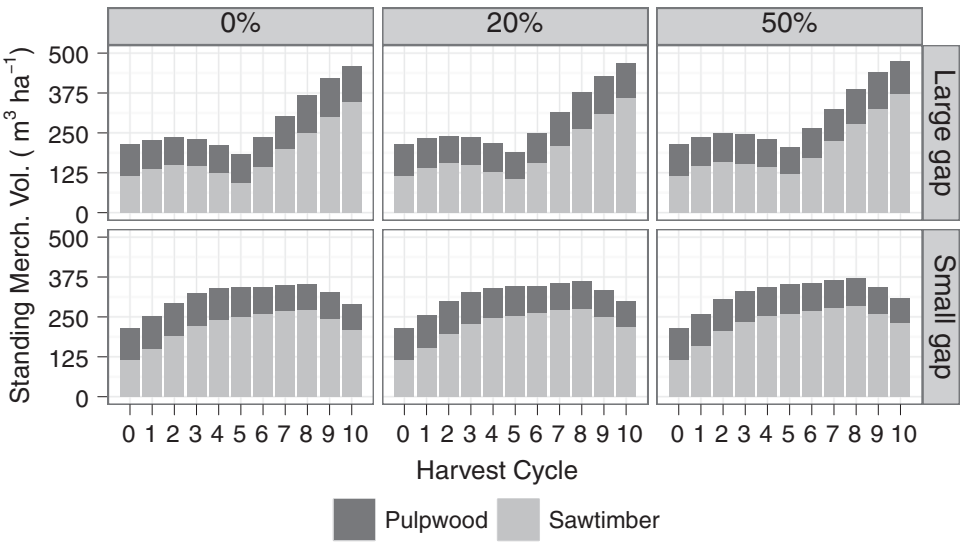

Fig. 9. Mean standing merchantable volume ($m^3\ ha^{-1}$) by 10 year harvest cycle for the large- and small-gap treatments with increased edge growth rates of 0%, 20%, and 50%.

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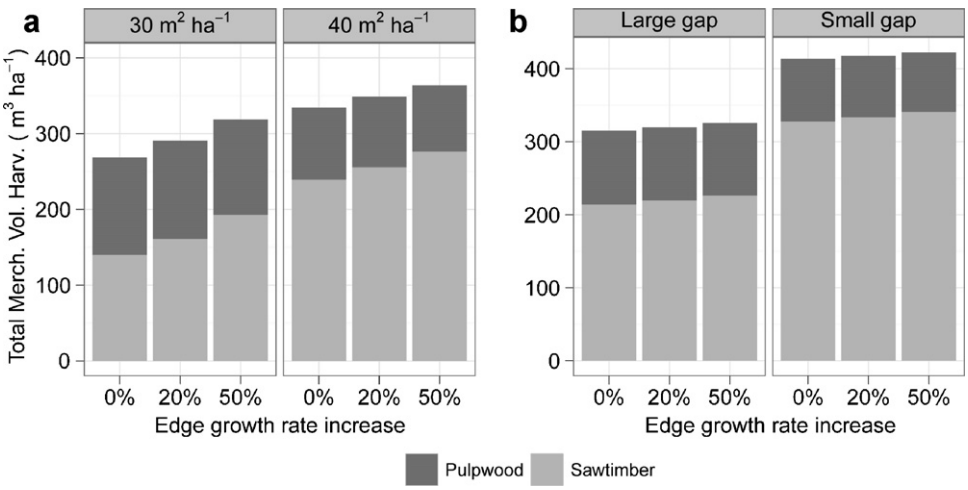

Fig. 10. Mean merchantable volume harvested ($m^3\ ha^{-1}$) at the end of the harvest rotation for (a) the 30 basal area plots (mean = $28.5 \pm 0.7\ m^2\ ha^{-1}$) and 40 basal area plots ($42.2 \pm 0.7\ m^2\ ha^{-1}$) and (b) the large- and small-gap treatments, each with increased edge growth rates of 0%, 20%, and 50%.

translate to a constant proportional increase in growth increments over time, and accordingly changes in predicted growth increment from multipliers will be greater in edge created early in the harvest rotation relative to edge created near the end (Fig. A.2).

Second, the implementation of the density mortality model and current limitations in modeling stagnation in FVS-NE caused the impact of edge effects on growth to diminish as plots approach the maximum sustainable basal area and equal zero when the plot reaches maximum basal area. This is a consequence of the underlying structure behind the FVS-NE growth model as well as the density dependent mortality model, designed to absorb gross stand basal area increment as plots approach site capacity (Dixon, 2002). The current formulation of the FVS-NE growth model estimates growth as a function of site index and social rank in the basal area distribution and has no term to modify growth based on proximity to maximum site capacity (Dixon and Keyser, 2008). Basal area subsequently tends to increase rapidly until it plateaus at the maximum sustainable basal area, which generally occurred early in the rotation. For example, it occurred during the third and fourth harvest cycles of the two plot subsets we examined (Fig. 5). Despite the large proportion of forest area in both treatments (60%–79%) which were in an edge condition for at least one harvest cycle over a rotation, the stand level differences in edge effect are a function of the small subset of plots which achieved an edge condition prior to reaching the maximum site capacity. Once site capacity was achieved, plot transitions into edge conditions and the resulting application of growth multipliers had no effect on ensuing stand development, which consequently resulted in an underestimation of the impacts of edge effects on long-term growth projections at the stand level.

Third, in both the small- and large-gap treatments, forest composition shifted towards a greater component of shade tolerant species. The mortality model implemented in FVS-NE distributes mortality as a function of shade tolerance and relative tree height (Dixon and Keyser, 2008), and all else being equal, favors killing small shade intolerant trees. In this analysis, edge growth multipliers were applied equally across species and diameter classes, while in reality, growth and survival responses in canopy gap-created edge forests vary with factors such as size, species, and shade tolerance (Arseneault et al., 2011; Coates et al., 2003; Menard et al., 2002). A simulation study by Banal et al. (2007) using the SORTIE model suggested while sapling basal area growth increased with increasing gap density, growth rates between species differed, with shade intolerants, such as red oak and white pine, exhibiting the greatest growth at high densities while species of intermediate shade tolerance, such as sugar maple and yellow birch, had the lowest increases. The differences in responses also increased with increasing gap density.

By customizing the growth modifiers to incorporate this variety in response, instead of using one multiplier for all species, it may be possible that gains in relative height growth for released shade intolerant species could potentially compensate for the increased likelihood of mortality due to lower shade tolerance. This would potentially enable the examination of changes in species composition patterns over time as a function of alternative harvest strategies and their respective edge creation patterns.

Nevertheless, despite these limitations we believe the results are reasonable given model parameters were based on reported ranges of edge effects distance (Banal et al., 2007; Menard et al., 2002; Olson, 2009), sapling and overstory growth rates in canopy-gap adjacent forests (Arseneault et al., 2011; Banal et al., 2007), and growth rate longevity (Pothier, 2002). Of the subset of edge growth rates we simulated in this experiment, the 20% increase scenario is likely the closest approximation of edge impacts on forest growth and yield. However, as previously mentioned, this approximation could be greatly improved if more specific approximations of growth rates by species and tree size in edge conditions could be estimated, either empirically or through improved causal understanding of growth and regeneration responses in canopy-gap adjacent forest.

4.2.1. Suggested modifications to FVS

This study clearly demonstrates a need to restructure the FVS model to allow incorporation of spatial data for growth projections of complex silvicultural systems. Aside from this, two key changes in model structure can be quickly made and they would greatly improve the FVS model's capabilities in simulating the composition, growth, and yield of complex stands. First, improvements in modeling growth stagnation in eastern variants are required. Some variants (such as the Central Rockies variant) have incorporated mechanisms to account for stagnation in diameter increment prediction. Integration of such mechanisms into other variants would likely improve the capabilities of modeling long-term dynamics in stands managed with complex silvicultural regimes. Second, a more flexible framework to specify and adjust maximum site capacity over time is necessary. Currently, the measure of maximum site capacity was fixed at the start of the simulation and derived from stand density index or maximum basal area and could not be modified over time to account for local increases in site capacity due to competitive release from adjacent harvest openings. The recent

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addition of the SETSITE keyword to the FVS model enables modifying site potential through time and, while currently untested, will likely create a more flexible framework and enhance FVS's capabilities to simulate forest development under a greater variety of complex silvicultural regimes.

4.3. Modeling implications for complex silvicultural systems

In the context of disturbance-based management, these complex harvesting regimes are designed at the strategic level to emulate many natural disturbance characteristics such as frequency, intensity, and other compositional, structural, and spatial patterns. At the tactical level, there are likely multiple possible variants of these silvicultural systems, each producing different patterns of edge over time, which could be utilized to meet those strategic level goals.

There is currently a paucity of research regarding the effects of gap creation on the growth and regeneration responses in gap-adjacent forest, and as such few studies exist to which our results can directly be compared. Some simple comparisons of factors such as gap size, density, and distribution, as well as broad insights into their impacts on gap-created edge dynamics can be inferred using mathematical approximations or from similar studies of harvest openings on forest patterns at either landscape- or stand-scales when available. For example, our observed differences in total edge creation between treatments were consistent with the landscape-level findings of Franklin and Forman (1987) in their study of the effects of number, size and arrangement of clearcut patches on the total length of border between forested and cutover areas. They reported, for a given area of removal, large cutting units create less border than small units, and low-density aggregated patterns, such as single-nucleus, four-nucleus, and progressive-parallel, create less border than high-density regular patterns, such as the dispersed patch model. These insights are of limited use when trying to compare alternative silvicultural system designs to determine which is most suitable to accomplish a variety of management objectives. Empirical studies alone cannot be used to determine optimal combinations of factors to meet a variety of management objectives. It would be infeasible to study the broad array of initial gap shapes, sizes, distributions, harvest frequencies, and their ensuing development over time under various methods of harvest allocation, such as multiple cohorts of initially irregular gap openings iteratively expanded asymmetrically over time. This study demonstrated how spatial simulation models could be useful tools to evaluate ensuing patterns in edge creation and longevity in response to the interacting effects of complex harvest regimes on the surrounding forest. Forest simulation models are increasingly being used to evaluate the long-term ecological and economical sustainability of complex silvicultural regimes. Given accumulating evidence regarding the effects of canopy gaps on adjacent forests and the potential impacts of edge effects on both short- and long-term forest development, incorporating patterns of edge creation over time into growth and yield projections will likely be necessary to accurately simulate forest dynamics when managed with complex silvicultural regimes.

Acknowledgments

We wish to thank Robert Wagner, Robert Seymour, and the University of Maine for access to the AFERP data for model calibration. We also thank Ralph Johnson and Patrick Zollner for help with designing the discrete space model, FVS simulations and reviewing an earlier version of this manuscript. Lastly, we wish to thank two anonymous reviewers for their insightful comments that greatly improved this article. This work was funded by Theme 1 of the Northern States Research Cooperative and a National Sciences and Engineering Council of Canada (NSERC) postgraduate research fellowship.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.ecolmodel.2012.04.003.

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